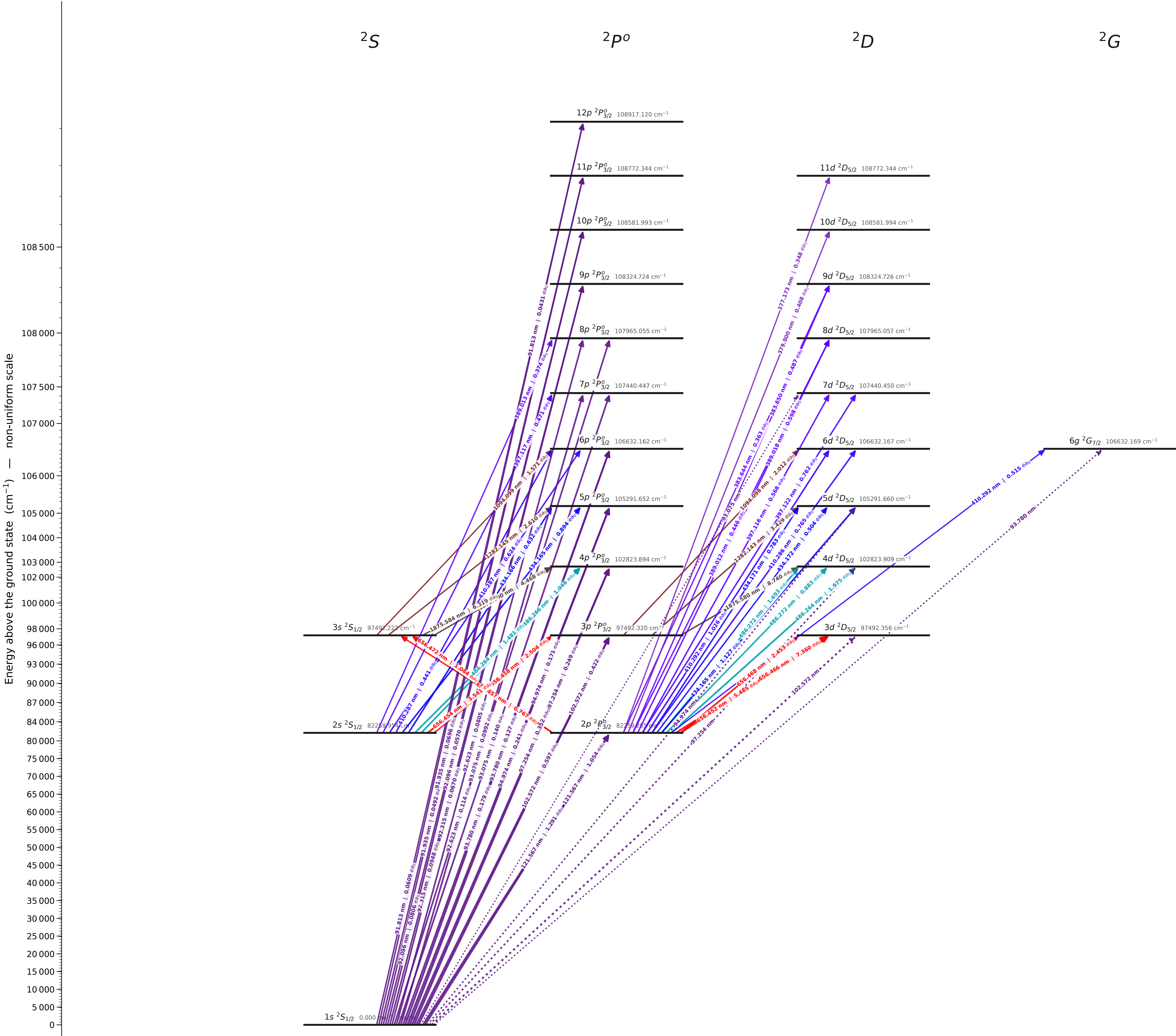


Hydrogen

H (H I). Strongest classified transitions below 2 μm (NIST): 42 levels, 66 lines.

H⁺ ionisation limit 109678.77 cm^{−1} = 13.59843 eV (91.175 nm)



Strongest transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	⟨J er J'⟩ (ea ₀)	A (s ^{−1})	Γ/2π up. (MHz)	branch (%)	Use
1s ² S _{1/2} - 6p ² P _{3/2} ^o	93.7803	-	3196.7518	0.179	1.973e+07	-	-	-
1s ² S _{1/2} - 6p ² P _{1/2} ^o	93.7803	-	3196.7514	0.127	1.973e+07	-	-	-
1s ² S _{1/2} - 5p ² P _{3/2} ^o	94.9743	-	3156.5643	0.241	3.438e+07	-	-	-
1s ² S _{1/2} - 5p ² P _{1/2} ^o	94.9743	-	3156.5636	0.171	3.438e+07	-	-	-
1s ² S _{1/2} - 4p ² P _{3/2} ^o	97.2537	-	3082.5828	0.352	6.818e+07	-	-	-
1s ² S _{1/2} - 4p ² P _{1/2} ^o	97.2537	-	3082.5814	0.249	6.818e+07	-	-	-
1s ² S _{1/2} - 3p ² P _{3/2} ^o	102.5722	-	2922.7462	0.597	1.672e+08	-	-	-
1s ² S _{1/2} - 3d ² D _{3/2}	102.5722	-	2922.7462	-	5.575e+07	-	-	-
1s ² S _{1/2} - 3p ² P _{1/2} ^o	102.5723	-	2922.7430	0.422	1.672e+08	-	-	-
1s ² S _{1/2} - 2p ² P _{3/2} ^o	121.5668	-	2466.0713	1.291	4.699e+08	-	-	-
1s ² S _{1/2} - 2p ² P _{1/2} ^o	121.5674	-	2466.0604	1.054	6.265e+08	-	-	-
2p ² P _{3/2} ^o - 4d ² D _{3/2}	486.2637	486.1279	616.5224	1.975	1.719e+07	-	-	-
2p ² P _{3/2} ^o - 3d ² D _{3/2}	656.4523	656.2710	456.6859	5.485	5.388e+07	-	-	-
2s ² S _{1/2} - 3p ² P _{3/2} ^o	656.4538	656.2725	456.6848	3.541	2.245e+07	-	-	-
2s ² S _{1/2} - 3p ² P _{1/2} ^o	656.4584	656.2771	456.6816	2.504	2.245e+07	-	-	-
2p ² P _{3/2} ^o - 3d ² D _{5/2}	656.4665	656.2852	456.6760	7.360	6.465e+07	-	-	-

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).

Arrow label: vacuum wavelength | reduced dipole matrix element |⟨J||er||J'⟩| in ea₀
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, n = 70.
Level caption: NIST energy and measured lifetime (where one exists).